COURSE SLIDES · WSQ

ISO 14064 Internal Auditor Training



WSQ Course Code: TGS-2025056357

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome and Course Orientation

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Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

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GENERAL TRAINER TEMPLATE

About the Trainer



Your Trainer

Complete this profile before class

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE
Principal Trainer and course developer

EXPERTISE
ISO internal auditing, GHG accounting, lifecycle assessment and evidence-based improvement

DELIVERS
WSQ professional and technical programmes

FOUNDER
Tertiary Infotech / Tertiary Courses

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LEARNER INTRODUCTION

Let's Know Each Other

1

You

Name, organisation and role.

2

Your footprint

One emission source or climate issue relevant to your work.

3

Your challenge

A recurring GHG data, reporting or reduction problem.

4

Your outcome

What successful learning looks like after two days.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time and complete attendance.

2. **Participate**
   Ask, challenge and contribute evidence.

3. **Respect**
   One conversation at a time; disagree professionally.

4. **Protect information**
   Use fictional or sanitised workplace data.

5. **Stay practical**
   Link every idea to a process or decision.

6. **Own the result**
   Verify your activity output before submission.

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SCHEDULE

Lesson Plan — Two Days

Day 1

- Day 1 · ISO 14064 architecture, internal-audit practice, GHG principles, scopes and boundary decisions

– 9:00-10:30 · administration, ISO family and audit principles

– 10:40-13:00 · audit planning and Activities 1-2

– 14:00-15:20 · evidence practice and Activity 3

– 15:30-18:00 · scopes, boundaries and Activities 4-5

Day 2

- Day 2 · Lifecycle inventory, quantification, uncertainty, benchmarking, reporting and audit close-out

– 9:00-10:30 · hotspots, mitigation and Activity 6

– 10:40-13:00 · lifecycle inventory and Activity 7

– 14:00-15:20 · quantification, uncertainty and Activities 8-9

– 15:30-16:00 · Activity 10 and wrap-up

– 16:00-18:00 · WA then Case Study

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Learning Outcomes

1

LO1

Develop process mappings and document GHG accounting reports aligned to ISO 14064 and international verification guidelines.

2

LO2

Define lifecycle boundaries and determine emission sources to identify carbon reduction opportunities in line with assessment principles.

3

LO3

Construct lifecycle inventory data for carbon footprint assessments using product configurations and scoped emission categories.

4

LO4

Determine uncertainty factors and evaluate product carbon performance against industry benchmarks.

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Knowledge Alignment · K1-K5

1

K1

ISO 14001, ISO 14064 and other relevant standards and guidelines

2

K2

Concepts and principles of carbon footprint assessments

3

K3

Product lifecycles and product configurations

4

K4

Types of emissions and corresponding sources

5

K5

Scopes of carbon footprints

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Knowledge Alignment · K6-K8

1

K6

Types of uncertainty factors

2

K7

Industry benchmarks for specific products' carbon footprints

3

K8

International guidelines and principles for carbon footprint assessments and documentation

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Ability Alignment · A1-A4

1

A1

Develop process mappings in accordance with product lifecycle requirements

2

A2

Define boundary conditions, assumptions and prioritisation for product lifecycles according to international guidelines and principles

3

A3

Construct lifecycle inventory data based on developed product lifecycle boundaries

4

A4

Assess carbon footprints of product lifecycles using carbon footprint assessment software tools

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Ability Alignment · A5-A8

1

A5

Determine uncertainty factors based on carbon footprint assessments

2

A6

Evaluate carbon footprint performance of selected products against industry benchmarks

3

A7

Determine emissions hot spots and propose potential carbon footprint reduction approaches

4

A8

Document results and prepare reports according to international guidelines and principles

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ASSESSMENT BRIEFING

Briefing for Assessment

1

Individual

No discussion; answer in your own words.

2

Open book

Use slides, Learner Guide and approved activity material only.

3

Secure

Place phones and other materials under the table; no recording.

4

Submission

Complete the provided document and upload to the LMS.

5

Competency

Every assessed K and A criterion must be Competent.

6

Rights

Re-assessment and appeal routes are available.

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FINAL ASSESSMENT

Assessment and Funding

Written Assessment

- 8 open-ended questions
- K1-K8 · 1 hour
- Open book

Case Study

- One coherent scenario
- A1-A8 · 1 hour
- Open book

Competency

- C in both instruments
- At least 75% attendance
- Funding criteria apply

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ON ASSESSMENT DAY · CONTROL POINTS

Assessment Readiness

1

Attendance

TRAQOM and assessment attendance are complete.

2

Instruments

Written Assessment and Case Study are completed individually.

3

Submission

Answers are uploaded to the LMS within the assessment window.

4

Record

The Assessment Summary Record is reviewed and signed.

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SELF-CONTAINED ACTIVITY PACKS

Access the Course Activities

Option A · Clone

- Open the repository link
- Copy the HTTPS clone URL
- Open the activities folder

Option B · ZIP

- GitHub · Code
- Download ZIP
- Extract and open activities/

Each folder

- Detailed README.md
- Working XLSX
- Printable PDF/checklist

Open the ISO 14064 Internal Auditor activity repository

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CONCEPT MODEL · DETAILED PROCEDURES ARE IN THE LEARNER GUIDE

Activity Evidence Architecture

1

Scenario

A realistic HarbourLight decision context with declared constraints.

2

Criteria

Applicable requirements, boundaries, assumptions and acceptance rules.

3

Evidence

Traceable records, calculations, interviews and professional judgement.

4

Conclusion

A decision-ready output tested against explicit acceptance criteria.

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https://lms-tms.tertiaryinfotech.com/

Course Portal

- Download the learner slides and Learner Guide.
- Open the complete Activities folder and working files.
- Complete and submit assessments only during the scheduled window.

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01

TOPIC 01

Introduction to ISO 14064 and Verification Principles

ISO 14064 family · ISO 14001 integration · accounting principles · audit and verification practice · reporting evidence



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K1 · K2 · K8 · A1 · A8 · LO1

Topic 01 · Core Model

1

Criteria

Audit against declared ISO, programme and organisational criteria—not auditor preference.

2

Evidence

Use verifiable records, observations, interviews and recalculation to support conclusions.

3

Risk

Focus effort where misstatement, omission or unsupported claims could affect users' decisions.

4

Reporting

Make scope, methods, limitations, findings and corrective actions transparent and traceable.

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K1 · K2 · K8 · A1 · A8 · LO1

ISO 14064 family and assurance architecture

1

Part 1

Organisation-level principles and requirements for inventory design, quantification and reporting.

2

Part 2

Project-level quantification, monitoring and reporting of emission reductions or removal enhancements.

3

Part 3

Requirements and guidance for validating and verifying GHG statements.

4

Internal audit

An organisation-led evaluation of readiness, controls and evidence; it does not itself create third-party assurance.

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Decision Lens

Apply in the organisation

- Name the applicable part and edition
- Separate organisation, project and product assertions
- Define intended users and assurance purpose
- Use authorised standards text for conformance

Avoid the trap

- ISO 14064 is not one undifferentiated certificate
- A project credit is not an organisational removal
- Internal audit does not replace independent verification
- Do not claim conformance from a checklist alone

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K1 · K2 · K8 · A1 · A8 · LO1

Purpose and architecture of ISO 14064-1:2018

1

Inventory design

Set purpose, organisational boundary, reporting boundary, base year and methods.

2

Development

Identify sources and sinks, select data and factors, quantify and consolidate.

3

Management

Assign roles, maintain information systems, assess uncertainty and review changes.

4

Reporting

Disclose boundaries, categories, methods, exclusions, uncertainty and performance transparently.

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Decision Lens

Apply in the organisation

- Trace each report claim to a controlled process
- Test the significance approach for indirect emissions
- Check base-year and recalculation rules
- Confirm management responsibility for the statement

Avoid the trap

- The 2018 edition is not a simple Scope 1-2-3 checklist
- Completeness still requires justified exclusions
- Verification readiness is not verified status
- A polished report can hide weak source controls

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K1 · K2 · K8 · A1 · A8 · LO1

ISO 14001 integration and the management-system link

1

Context

Climate, stakeholder and compliance issues influence EMS and GHG priorities.

2

Planning

Environmental aspects, obligations, risks and objectives can incorporate GHG sources and targets.

3

Support and operation

Competence, communication, document control and operational controls support inventory reliability.

4

Evaluation and improvement

Monitoring, internal audit, management review and corrective action create a repeatable control cycle.

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Decision Lens

Apply in the organisation

- Reuse governance without merging criteria
- Map GHG data owners into EMS roles
- Connect findings to corrective action
- Escalate material climate and claims risks

Avoid the trap

- ISO 14001 certification does not prove inventory accuracy
- GHG boundaries can differ from EMS boundaries
- One audit plan may not cover both standards fully
- Avoid duplicate uncontrolled records

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K1 · K2 · K8 · A1 · A8 · LO1

Related standards and frameworks

1

ISO 14065 and 14066

Requirements for validation/verification bodies and competence of GHG teams.

2

ISO 14067

Product carbon-footprint principles and requirements within lifecycle assessment.

3

ISO 19011

Guidance for audit programmes, risk-based planning, evidence and auditor competence.

4

GHG Protocol

Widely used corporate, Scope 2 and value-chain accounting guidance that can complement declared ISO criteria.

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Decision Lens

Apply in the organisation

- Create a criteria crosswalk
- Record definitions that differ
- Use programme rules when contractually applicable
- Keep the controlling criteria explicit

Avoid the trap

- Related does not mean interchangeable
- Scope labels and ISO categories are not identical
- Product and organisation footprints answer different questions
- A crosswalk needs expert judgement

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K1 · K2 · K8 · A1 · A8 · LO1

GHG accounting principles as audit tests

1

Relevance

Boundaries and information serve intended users and decisions.

2

Completeness

All significant sources and sinks are included or exclusions are justified.

3

Consistency

Methods support comparison over time and changes are disclosed.

4

Accuracy and transparency

Bias and uncertainty are reduced and assumptions, evidence and limitations remain visible.

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Decision Lens

Apply in the organisation

- Turn every principle into an audit question
- Use principles when prescriptive rules are silent
- Seek evidence of trade-off decisions
- Apply professional scepticism to favourable claims

Avoid the trap

- Principles interact and can conflict
- Completeness does not mean perfect data
- Consistency does not freeze weak methods
- Transparency does not cure avoidable error

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K1 · K2 · K8 · A1 · A8 · LO1

Verification planning: scope, criteria and assurance

1

Objectives

State what the engagement must conclude and for whom.

2

Scope

Define statement, entity, locations, period, categories and activities covered.

3

Criteria

Name the standard, programme rules, calculation policies and materiality basis.

4

Assurance

Limited and reasonable assurance require different evidence depth and conclusion forms.

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Decision Lens

Apply in the organisation

- Resolve ambiguous criteria before testing
- Link risk assessment to evidence effort
- Document changes to scope
- Align competence and resources to complexity

Avoid the trap

- More samples do not fix unclear criteria
- Materiality is not only a percentage
- Limited assurance is not low quality
- Internal audit should not mimic an external opinion

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K1 · K2 · K8 · A1 · A8 · LO1

Risk-based internal audit process

1

Programme

Prioritise audits by significance, change, prior findings and stakeholder risk.

2

Plan

Set objectives, scope, criteria, team, independence, timetable and sampling strategy.

3

Conduct

Open, interview, observe, inspect, recalculate, corroborate and communicate emerging issues.

4

Close and follow up

Report findings, agree responsibilities, verify correction and test effectiveness.

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Decision Lens

Apply in the organisation

- Plan backward from required conclusions
- Maintain independence from inventory preparation
- Use daily team calibration
- Track findings to verified closure

Avoid the trap

- An audit is not an inspection tour
- Interview statements need corroboration
- Prescribing solutions can compromise independence
- Closing a task is not proving effectiveness

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K1 · K2 · K8 · A1 · A8 · LO1

Objective evidence, sampling and reporting

1

Evidence types

Records, observations, interviews, confirmations, analytics and reperformance provide different strengths.

2

Sampling

Select high-value, high-risk, unusual and representative items using a documented rationale.

3

Triangulation

Corroborate important conclusions across independent evidence sources.

4

Report trail

Link criterion, evidence, evaluation, finding, owner, correction and follow-up status.

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Decision Lens

Apply in the organisation

- Trace top-down and bottom-up
- Record population and sample logic
- Retest corrected calculations
- Distinguish fact, judgement and recommendation

Avoid the trap

- A large sample can still be biased
- One document may be internally consistent but false
- Absence of evidence is not always evidence of absence
- Avoid vague findings such as documentation weak

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Activities · Topic 1

Activity 1

- Map the GHG System and Audit Criteria
- K1 · K2 · K8 · A1 · A8 · LO1
- 45 minutes

Activity 2

- Plan a Risk-Based ISO 14064 Internal Audit
- K1 · K8 · A8 · LO1
- 45 minutes

Activity 3

- Conduct Evidence Interviews and a Reporting Gap Review
- K2 · K8 · A8 · LO1
- 50 minutes

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K1 · K2 · K8 · A1 · A8 · LO1 · 45 minutes

Map the GHG System and Audit Criteria

ACTIVITY 1

Learners join HarbourLight Electronics' internal audit team and turn fragmented GHG processes into a criterion-linked system map.

You'll build

A GHG process map, criteria crosswalk, document request list and audit-ready reporting index.

Tools: Scenario dossier · criteria cards · process-map workbook · report checklist

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K1 · K2 · K8 · A1 · A8 · LO1 · CONCEPT VIEW

Activity 1 · Evidence Model

1 **Criteria**
ISO and programme crosswalk

2 **Process**
GHG reporting-system map

3 **Requests**
Risk-linked evidence list

4 **Index**
Audit-ready reporting trail

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ACTIVITY 1 · ACCEPTANCE CHECK

Map the GHG System and Audit Criteria

✅ Test it

Every assertion is assigned to declared criteria; the process map shows all major data and approval hand-offs; the reporting index links requirements to retrievable evidence.

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K1 · K8 · A8 · LO1 · 45 minutes

Plan a Risk-Based ISO 14064 Internal Audit

ACTIVITY 2

Learners convert HarbourLight's audit mandate into a defensible plan that focuses effort on material misstatement and claims risk.

You'll build

An audit charter, risk assessment, sampling plan, interview schedule and opening-meeting agenda.

Tools: Audit mandate · prior-finding summary · population listings · planning workbook

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K1 · K8 · A8 · LO1 · CONCEPT VIEW

Activity 2 · Evidence Model

1 **Charter**
Objectives, scope and criteria

2 **Risk**
Material-misstatement screen

3 **Sampling**
Evidence-selection rationale

4 **Timetable**
Independent fieldwork plan

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ACTIVITY 2 · ACCEPTANCE CHECK

Plan a Risk-Based ISO 14064 Internal Audit

✅ Test it

The plan links risks to procedures and samples, defines independence and competence, covers both statements without scope ambiguity and can support the planned conclusions.

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K2 · K8 · A8 · LO1 · 50 minutes

Conduct Evidence Interviews and a Reporting Gap Review

ACTIVITY 3

Learners conduct simulated interviews and document testing across HarbourLight's GHG reporting process.

You'll build

Interview records, evidence trails, recalculation samples, criterion-based findings and an issues briefing.

Tools: Role cards · evidence pack · draft report · finding workbook

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K2 · K8 · A8 · LO1 · CONCEPT VIEW

Activity 3 · Evidence Model

1. **Interviews**
   Corroborated process evidence

2. **Trace**
   Source-to-report lineage

3. **Recalculation**
   Independent model tests

4. **Findings**
   Criterion-evidence evaluation

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ACTIVITY 3 · ACCEPTANCE CHECK

Conduct Evidence Interviews and a Reporting Gap Review

✅ Test it

Important interview claims are corroborated; recalculations are reproducible; findings link criterion, evidence and evaluation; fact and judgement remain clearly separated.

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K1 · K2 · K8 · A1 · A8 · LO1 · TOPIC CHECK

Recap · ISO 14064 and Verification Principles

1

Explain

Audit against declared ISO, programme and organisational criteria—not auditor preference.

2

Apply

Translate ISO 14064 and related criteria into a process map and auditable requirements register.

3

Demonstrate

Define audit objectives, scope, criteria, independence, risks, sampling and timetable for a two-day internal audit.

4

Evidence

Retain the completed activity artefacts and review feedback.

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02

TOPIC 02

GHG Sources, Boundaries and Mitigation Activities

Scope 1, 2 and 3 · sources, sinks and reservoirs · organisational and lifecycle boundaries · hotspots · mitigation hierarchy



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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Topic 02 · Core Model

1

Boundary

Define the organisation, product system, lifecycle stages, functional unit and exclusions before calculating.

2

Classification

Distinguish direct emissions, imported energy and other indirect value-chain sources consistently.

3

Significance

Prioritise indirect sources using magnitude, influence, risk, data quality and stakeholder interest.

4

Mitigation

Target material hotspots while checking cost, feasibility, burden shifting and evidence quality.

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Scope 1, 2 and 3 alongside ISO reporting categories

1

Scope 1

Direct emissions from sources owned or controlled by the organisation.

2

Scope 2

Indirect emissions from generation of purchased electricity, steam, heat or cooling.

3

Scope 3

Other indirect upstream and downstream value-chain emissions.

4

ISO view

Direct emissions plus five indirect-emission categories organised around imported energy, transport, products and other sources.

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Decision Lens

Apply in the organisation

- Classify by ownership, control and activity
- Map every source to both reporting views where needed
- Keep location- and market-based Scope 2 distinct
- Test for double counting across categories

Avoid the trap

- Outsourcing does not remove significance
- A supplier invoice is not automatically Scope 3 evidence
- Scope classification does not prove completeness
- Do not force ISO categories into GHG Protocol labels silently

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Greenhouse gases, GWP and CO2e

1

Gas inventory

Carbon dioxide, methane, nitrous oxide and covered fluorinated gases require source-specific consideration.

2

Global warming potential

GWP converts gas mass to a common CO2-equivalent basis over a declared time horizon.

3

Source basis

The selected IPCC assessment or programme basis must be versioned and applied consistently.

4

Audit trail

Gas mass, factor, GWP value and CO2e result should remain separately traceable.

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Decision Lens

Apply in the organisation

- Check all relevant gases
- Reperform factor and GWP multiplication
- Test consistency across reporting years
- Disclose changes to GWP basis

Avoid the trap

- CO2e is a convention, not measured carbon dioxide
- GWP tables change with scientific assessment
- Refrigerant blends require correct composition
- Rounding can hide material gas-level errors

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Sources, sinks, reservoirs, removals and credits

1

Source

A process or activity that releases a GHG to the atmosphere.

2

Sink

A process that removes a GHG from the atmosphere.

3

Reservoir

A physical unit or component that stores a GHG or carbon stock.

4

Credit

An external unit representing a project outcome; it is not automatically an organisational removal.

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Decision Lens

Apply in the organisation

- Map physical processes before accounting labels
- Require monitoring for removal claims
- Evaluate reversal and permanence
- Report gross inventory and external credits separately

Avoid the trap

- Trees on site do not guarantee a quantified removal
- Avoided emissions differ from removals
- Biogenic CO2 needs declared treatment
- Purchased credits cannot erase source records

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Organisational boundary decisions

1

Equity share

Account according to the organisation's economic interest.

2

Financial control

Include operations governed through financial and operating policies.

3

Operational control

Include operations where the organisation can introduce and implement operating policies.

4

Evidence

Ownership documents, leases, contracts, board rights and operating authority support the conclusion.

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Decision Lens

Apply in the organisation

- Map every entity and operation
- Apply one consolidation approach consistently
- Review joint arrangements and leases
- Reassess after acquisitions or divestments

Avoid the trap

- Legal ownership and control can differ
- A boundary choice can change trends materially
- Contractors may remain significant indirect sources
- Do not cherry-pick low-emission operations

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Reporting boundary and significant indirect emissions

1

Inventory universe

Start with all plausible direct and indirect sources across operations and value chain.

2

Significance criteria

Use magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.

3

Decision log

Retain scores, evidence, inclusion decisions, exclusions, owners and review dates.

4

Change control

Reassess significance when products, suppliers, sites, regulation or user needs change.

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Decision Lens

Apply in the organisation

- Challenge low-data categories
- Check procurement and logistics completeness
- Document thresholds before seeing results
- Report significant exclusions transparently

Avoid the trap

- Data difficulty is not an exclusion criterion
- Magnitude alone can miss claims risk
- Spend-based screening is not final quantification
- A significance score is judgement, not proof

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Product lifecycle boundaries and assumptions

1

Goal and scope

State intended application, audience, product configuration and decision context.

2

Functional unit

Define the quantified function and performance basis for consistent comparison.

3

System boundary

Specify cradle-to-gate, cradle-to-grave or other included lifecycle stages.

4

Rules

Document cut-offs, allocation, recycled content, use profile, lifetime and end-of-life assumptions.

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Decision Lens

Apply in the organisation

- Draw the physical process map
- Record included and excluded flows
- Test sensitivity to major assumptions
- Review comparability before benchmarking

Avoid the trap

- A product unit is not always a functional unit
- Cut-offs can hide hotspots
- Recycling can be double credited
- Use-phase assumptions often dominate results

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Hotspot analysis and prioritisation

1

Magnitude

Rank contribution by lifecycle stage, source, gas and product component.

2

Influence

Assess control, supplier leverage, design authority and customer-use influence.

3

Confidence

Rate data quality and test whether uncertainty could change the ranking.

4

Decision

Combine emissions, feasibility, cost, timing, co-impact and risk into transparent priorities.

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Decision Lens

Apply in the organisation

- Use contribution and Pareto views
- Run sensitivity on weak high-impact inputs
- Separate screening from investment-grade analysis
- Document why lower-ranked actions proceed

Avoid the trap

- Largest is not always most controllable
- Spend hotspots can reflect price, not mass
- A precise rank can be false precision
- Do not ignore small high-risk fugitive sources

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Mitigation hierarchy and audit tests

1

Avoid

Remove unnecessary activity, demand or material before optimising it.

2

Reduce

Improve efficiency, yield, maintenance, logistics and process control.

3

Substitute

Change fuel, material, technology, design or service model.

4

Residual

Treat credits or removals separately and only after credible internal and value-chain action.

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Decision Lens

Apply in the organisation

- Quantify expected and realised abatement
- Check burden shifting across lifecycle stages
- Assign owners, milestones and evidence
- Verify claims against inventory results

Avoid the trap

- Forecast savings are not realised reductions
- Intensity can fall while absolute emissions rise
- Renewable certificates need claim controls
- Offsets are not operational reduction

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Activities · Topic 2

Activity 4

- Define Organisational and Product Lifecycle Boundaries
- K3 · K5 · A1 · A2 · LO2
- 50 minutes

Activity 5

- Build the GHG Source, Sink and Scope Register
- K4 · K5 · A2 · A7 · LO2
- 45 minutes

Activity 6

- Diagnose Hotspots and Design a Mitigation Roadmap
- K2 · K3 · K4 · A7 · LO2
- 45 minutes

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K3 · K5 · A1 · A2 · LO2 · 50 minutes

Define Organisational and Product Lifecycle Boundaries

ACTIVITY 4

Learners resolve ownership, control, outsourced activity, functional-unit and lifecycle boundary decisions before quantification.

You'll build

An organisational boundary diagram, lifecycle process map, functional unit, system-boundary statement and assumptions log.

Tools: Group structure · product bill of materials · logistics facts · boundary workbook

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K3 · K5 · A1 · A2 · LO2 · CONCEPT VIEW

Activity 4 · Evidence Model

1 **Ownership**
Entity and control view

2 **Functional unit**
Comparable product basis

3 **Lifecycle**
Included stages and flows

4 **Assumptions**
Transparent boundary decisions

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ACTIVITY 4 · ACCEPTANCE CHECK

Define Organisational and Product Lifecycle Boundaries

✅ Test it

Every named operation and lifecycle stage is addressed; the functional unit is comparable and measurable; exclusions and assumptions have transparent rationale, evidence and ownership.

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K4 · K5 · A2 · A7 · LO2 · 45 minutes

Build the GHG Source, Sink and Scope Register

ACTIVITY 5

Learners perform a structured completeness review of HarbourLight's operations and Lumina 400 value chain.

You'll build

A source/sink register, Scope and ISO-category map, significance scoring model and data-owner matrix.

Tools: Process narrative · ledger categories · facility records · significance workbook

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K4 · K5 · A2 · A7 · LO2 · CONCEPT VIEW

Activity 5 · Evidence Model

1. **Register**
   Sources, sinks and gases

2. **Classification**
   Scope and ISO categories

3. **Significance**
   Evidence-based prioritisation

4. **Ownership**
   Data stewards and controls

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ACTIVITY 5 · ACCEPTANCE CHECK

Build the GHG Source, Sink and Scope Register

✅ Test it

All scenario sources and gases are addressed; Scope and ISO categories are consistent; removals and credits remain separate; significance and ownership decisions are evidence-based.

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K2 · K3 · K4 · A7 · LO2 · 45 minutes

Diagnose Hotspots and Design a Mitigation Roadmap

ACTIVITY 6

Learners convert an initial screening footprint into a practical 24-month reduction roadmap for HarbourLight and the Lumina 400.

You'll build

A hotspot dashboard, influence-confidence matrix, mitigation options, prioritisation scorecard and phased roadmap.

Tools: Screening inventory · option fact cards · lifecycle map · roadmap workbook

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K2 · K3 · K4 · A7 · LO2 · CONCEPT VIEW

Activity 6 · Evidence Model

1 **Hotspots**
Magnitude and contribution

2 **Influence**
Control and leverage

3 **Sensitivity**
Decision robustness

4 **Roadmap**
Phased mitigation evidence

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ACTIVITY 6 · ACCEPTANCE CHECK

Diagnose Hotspots and Design a Mitigation Roadmap

✅ Test it

Priorities reflect magnitude, influence and uncertainty; selected actions avoid undisclosed burden shifts; milestones, owners and evidence requirements are explicit; no forecast is presented as achieved reduction.

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K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2 · TOPIC CHECK

Recap · Sources, Boundaries and Mitigation

1

Explain

Define the organisation, product system, lifecycle stages, functional unit and exclusions before calculating.

2

Apply

Apply consolidation, reporting-boundary and lifecycle rules to HarbourLight's organisation and Lumina 400 product.

3

Demonstrate

Identify gases, sources, sinks and reservoirs; classify them consistently and assess significant indirect emissions.

4

Evidence

Retain the completed activity artefacts and review feedback.

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DAY 2

Reduction Strategy → Carbon Strategy

03

TOPIC 03

Lifecycle Assessment and Carbon Footprint Quantification

Lifecycle phases · process mapping · lifecycle inventory quantification models and software · inventory controls



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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Topic 03 · Core Model

1

Functional unit

Anchor comparisons to the same product function, performance and service life.

2

Inventory

Collect traceable material, energy, transport, waste and emission data for every included unit process.

3

Model

Connect foreground activity data to documented background factors and declared GWP values.

4

Control

Version factors, protect formulas, reconcile totals and preserve the audit trail from source to report.

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

LCA phases and product configurations

1

Goal and scope

Define purpose, audience, functional unit, product variants and system boundary.

2

Inventory

Compile quantified inputs and outputs for each unit process.

3

Impact assessment

Apply declared characterisation methods such as 100-year GWP.

4

Interpretation

Evaluate completeness, sensitivity, consistency, hotspots and limitations.

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Decision Lens

Apply in the organisation

- Keep configuration changes visible
- Compare products on equivalent function
- Iterate when data gaps affect the goal
- Separate measured findings from scenario results

Avoid the trap

- LCA is iterative, not a one-pass calculation
- A lighter product may perform differently
- Database completeness does not guarantee relevance
- Do not over-generalise one configuration

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Process mapping for lifecycle evidence

1

Unit processes

Break the lifecycle into decision-relevant transformation, transport, use and end-of-life stages.

2

Inputs

Capture materials, energy, water, auxiliaries, packaging and services.

3

Outputs

Record product, co-products, emissions, wastewater, solid waste and recyclable flows.

4

Interfaces

Assign data owners, meters, invoices, supplier records and model links at each boundary.

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Decision Lens

Apply in the organisation

- Walk the process with operations staff
- Reconcile mass and energy where possible
- Mark estimates and missing evidence
- Use the map to drive sampling

Avoid the trap

- A flowchart without quantities is not an inventory
- Supplier processes may be hidden
- Waste leaving site is not impact-free
- Do not mix product and organisational boundaries

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Lifecycle inventory data architecture

1

Foreground data

Company- or supplier-specific activity data for processes under direct influence.

2

Background data

Database processes and factors for upstream or downstream activities.

3

Metadata

Source, owner, period, geography, technology, unit, uncertainty and approval accompany each value.

4

Balance

Mass, energy, production and financial reconciliations reveal omissions and unit errors.

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Decision Lens

Apply in the organisation

- Create a data dictionary
- Use stable IDs for processes and factors
- Retain raw and transformed values
- Reconcile totals to operational records

Avoid the trap

- Primary data can still be poor
- Generic data can be appropriate but must be disclosed
- Units are a frequent large-error source
- Missing metadata destroys auditability

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Data hierarchy and emission-factor selection

1

Direct measurement

Calibrated meters or continuous monitoring can support source-specific results.

2

Activity-based

Physical activity multiplied by a relevant documented factor is common and transparent.

3

Supplier-specific

Verified product or facility data can improve relevance when boundaries align.

4

Proxy or spend

Screening estimates are useful when labelled and improved for material sources.

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Decision Lens

Apply in the organisation

- Match geography, technology and vintage
- Record factor units and GWP inclusion
- Evaluate supplier boundary comparability
- Use conservative proxies where justified

Avoid the trap

- Newest is not always most representative
- A national factor may mask regional grids
- Supplier claims may net credits
- Factor precision can exceed data quality

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Quantification model and calculation controls

1

Equation

Activity data × emission factor × required conversion or GWP produces the declared result.

2

Allocation

Shared-process burdens need a justified physical, economic or other rule.

3

Aggregation

Gas, source, process, stage, scope and category totals should reconcile.

4

Controls

Locked formulas, validation rules, peer recalculation and version logs reduce silent error.

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Decision Lens

Apply in the organisation

- Standardise units before multiplication
- Reperform high-risk rows independently
- Test sign and decimal errors
- Preserve bridges between versions

Avoid the trap

- A balanced total can contain offsetting errors
- Allocation can change conclusions
- Do not net credits into gross emissions
- Spreadsheet protection is not review

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Carbon-footprint software workflow

1

Set up

Define functional unit, product configuration, system boundary and impact method.

2

Model

Create unit processes, enter foreground data and link controlled background datasets.

3

Analyse

Run totals, contribution analysis, sensitivity, scenarios and data-quality review.

4

Export

Preserve model version, datasets, settings, results and report tables for independent review.

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Decision Lens

Apply in the organisation

- Validate one manual calculation
- Review every high-contribution model link
- Save assumptions outside the software
- Re-run after controlled corrections

Avoid the trap

- Software does not decide the boundary
- Database names can hide different technologies
- A colourful Sankey is not assurance
- Model updates can change results silently

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Inventory information systems and internal controls

1

Ownership

Every source, factor and report output has accountable preparation and review roles.

2

Access

Permissions separate input, method approval, review and publication responsibilities.

3

Change

Version control records who changed what, why, when and with which approval.

4

Retention

Source records, calculations, reviews and approvals remain retrievable for the defined period.

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Decision Lens

Apply in the organisation

- Map data lineage end to end
- Test preventive and detective controls
- Inspect exception logs
- Sample user access and approvals

Avoid the trap

- Sustainability should not own every source
- Email approval may not control the file
- Automation can scale errors
- Backups do not prove record integrity

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Base year, recalculation and verification-ready evidence

1

Base year

Select a representative period with sufficient data and state why it is suitable.

2

Recalculation

Define triggers for structural change, methodology change and significant error.

3

Evidence pack

Index boundary decisions, source records, factors, calculations, controls, uncertainty and approvals.

4

Management review

Resolve findings and accept residual limitations before the statement is issued.

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Decision Lens

Apply in the organisation

- Publish the recalculation policy in advance
- Build an audit-ready evidence index
- Retain old and new method bridges
- Require formal inventory approval

Avoid the trap

- Ordinary performance change is not recalculation
- Acquisitions can break comparability
- A data room without an index is inefficient
- Management approval does not replace verification

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Activities · Topic 3

Activity 7

- Construct the Lumina 400 Lifecycle Inventory
- K3 · K4 · K5 · A1 · A3 · LO3
- 50 minutes

Activity 8

- Quantify and Audit the Product Carbon Footprint Model
- K2 · K4 · K5 · A3 · A4 · LO3
- 55 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

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K3 · K4 · K5 · A1 · A3 · LO3 · 50 minutes

Construct the Lumina 400 Lifecycle Inventory

ACTIVITY 7

Learners translate the Lumina 400 process map into a unit-process inventory with auditable data sources and balances.

You'll build

An LCI data dictionary, unit-process register, material and energy inventory, data-quality ratings and gap plan.

Tools: Bill of materials · production and supplier data · transport facts · LCI workbook

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K3 · K4 · K5 · A1 · A3 · LO3 · CONCEPT VIEW

Activity 7 · Evidence Model

1 **Unit processes**
Boundary-aligned process set

2 **Inventory**
Material and energy flows

3 **Metadata**
Source, unit and quality

4 **Balance**
Reconciliation and gaps

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ACTIVITY 7 · ACCEPTANCE CHECK

Construct the Lumina 400 Lifecycle Inventory

✅ Test it

Every included unit process has traceable inputs and outputs; metadata and quality ratings are complete; totals reconcile where possible; high-impact gaps have owned improvement actions.

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K2 · K4 · K5 · A3 · A4 · LO3 · 55 minutes

Quantify and Audit the Product Carbon Footprint Model

ACTIVITY 8

Learners build a tool-neutral calculation model, run contribution analysis and audit the trail from source record to reported footprint.

You'll build

A factor register, calculated footprint, software setup record, contribution analysis, audit tests and corrected result bridge.

Tools: LCI output · factor catalogue · calculation workbook · model review checklist

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K2 · K4 · K5 · A3 · A4 · LO3 · CONCEPT VIEW

Activity 8 · Evidence Model

1

Factor control

Source, version and units

2

Model

Reproducible footprint

3

Contributions

Decision-relevant hotspots

4

Corrections

Traceable audit bridge

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ACTIVITY 8 · ACCEPTANCE CHECK

Quantify and Audit the Product Carbon Footprint Model

✅ Test it

The model is reproducible, units and factors are controlled, seeded defects are resolved, contribution results reconcile, and the correction bridge preserves a complete audit trail.

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K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3 · TOPIC CHECK

Recap · Lifecycle and Footprint Quantification

1

Explain

Anchor comparisons to the same product function, performance and service life.

2

Apply

Build a complete, metadata-rich lifecycle inventory from the approved product boundary and configuration.

3

Demonstrate

Use a controlled spreadsheet/software workflow to quantify the Lumina 400 footprint and test its model integrity.

4

Evidence

Retain the completed activity artefacts and review feedback.

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04

TOPIC 04

Emission Calculations, Uncertainty and Performance Evaluation

Calculations and removals · uncertainty · sensitivity · product benchmarks · trends · audit findings and follow-up



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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Topic 04 · Core Model

1

Calculation

Use compatible units, appropriate factors, declared GWP basis and transparent allocation.

2

Uncertainty

Assess activity data, factors, models, sampling and boundary judgement—not only spreadsheet arithmetic.

3

Benchmark

Compare only after aligning functional unit, system boundary, geography, technology, vintage and method.

4

Finding

Link criterion, objective evidence, evaluation and consequence; verify correction and effectiveness.

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Calculating stationary, mobile, process and fugitive emissions

1

Stationary combustion

Fuel use, calorific basis, oxidation and gas-specific factors support boiler and generator emissions.

2

Mobile combustion

Fuel-based methods are generally stronger than distance estimates when reliable consumption data exists.

3

Process

Material balance, chemistry or direct monitoring may quantify non-combustion releases.

4

Fugitive

Mass-balance, service records, leak detection and equipment inventories support refrigerant and gas losses.

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Decision Lens

Apply in the organisation

- Confirm activity units and period
- Check factor scope and oxidation assumptions
- Reconcile purchases, stocks and consumption
- Sample maintenance and loss records

Avoid the trap

- Invoice totals may include non-reporting periods
- Distance models hide load and fuel variation
- Process factors may be technology-specific
- Top-up is not always equal to annual leakage

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Purchased-energy accounting and renewable claims

1

Location-based

Use grid-average factors reflecting physical electricity consumption in the relevant geography.

2

Market-based

Use qualifying contractual instruments under declared quality and allocation rules.

3

Steam, heat and cooling

Identify supplier, energy content, losses and appropriate generation factors.

4

Controls

Check instrument ownership, geography, vintage, exclusivity, cancellation or retirement and residual mix.

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Decision Lens

Apply in the organisation

- Report both Scope 2 views where criteria require
- Reconcile consumption to contracts
- Prevent double allocation of certificates
- Keep physical efficiency visible

Avoid the trap

- Zero-emission claims need criteria
- Certificates do not change onsite consumption
- Supplier-specific factors can embed offsets
- Renewable procurement is not the same as avoided emissions

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Removals, avoided emissions and net presentation

1

Organisational removal

A quantified atmospheric removal attributable within the declared inventory boundary.

2

Project reduction

A change relative to a baseline under project-level criteria.

3

Avoided emissions

A comparative consequence outside the gross inventory and dependent on a counterfactual.

4

Net claim

A communication that must disclose gross emissions, reductions, removals, credits, boundary and period.

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Decision Lens

Apply in the organisation

- Test ownership and boundary
- Review baseline and permanence
- Report comparative results separately
- Challenge claim wording against evidence

Avoid the trap

- A renewable product does not create negative Scope 3 automatically
- Credits are not company removals
- Counterfactuals are uncertain
- Net numbers can conceal rising gross emissions

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Uncertainty taxonomy and data-quality evaluation

1

Activity data

Meter accuracy, estimation, completeness, temporal match and sampling affect inputs.

2

Emission factors

Geography, technology, vintage, representativeness and statistical range affect conversion.

3

Model

Allocation, system boundary, use profile, lifetime and end-of-life assumptions affect structure.

4

Scenario and judgement

Future conditions, significance, exclusions and expert choices may alter conclusions.

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Decision Lens

Apply in the organisation

- Rate reliability and representativeness
- Separate known error from uncertainty
- Prioritise improvement by decision impact
- Disclose residual uncertainty

Avoid the trap

- A single percentage can hide different uncertainty types
- Low uncertainty does not mean no bias
- Qualitative assessment can still be rigorous
- Do not claim confidence beyond evidence

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Sensitivity, scenarios and quantitative uncertainty

1

Sensitivity

Change one influential input or assumption to test its effect on the result.

2

Scenario

Compare coherent alternative futures, technologies, boundaries or use profiles.

3

Propagation

Combine input distributions analytically or through Monte Carlo simulation where justified.

4

Decision threshold

Ask whether uncertainty could change ranking, conformance, target status or investment choice.

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Decision Lens

Apply in the organisation

- Start with high-impact weak inputs
- Define plausible ranges and sources
- Report distributions and assumptions
- Tie analysis to a decision

Avoid the trap

- Wide ranges are not a substitute for better data
- Inputs may be correlated
- Monte Carlo precision can be misleading
- Scenario outcomes are not forecasts

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Product carbon-footprint benchmarking

1

Comparable unit

Align functional unit, performance, service life and product configuration.

2

Comparable boundary

Match lifecycle stages, cut-offs, allocation, recycling and use assumptions.

3

Comparable method

Align GWP basis, database, geography, technology and assessment vintage.

4

Evidence source

Use verified EPDs, product category rules, industry datasets and peer-reviewed studies with stated limitations.

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Decision Lens

Apply in the organisation

- Build a comparability screen before ranking
- Normalise only when defensible
- Show range and quartile, not one magic number
- Verify claims before marketing use

Avoid the trap

- Different PCRs may prevent comparison
- Regional grid differences can dominate
- Best-in-class data may be selective
- A lower number does not prove overall environmental superiority

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Performance over time and causal interpretation

1

Absolute

Total emissions show atmospheric contribution and scale effects.

2

Intensity

Emissions per unit of output or function show efficiency but depend on denominator quality.

3

Bridge

Separate activity, structure, factor, method and genuine reduction effects between periods.

4

Control chart

Monitor trends, thresholds and anomalies without confusing normal variation with improvement.

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Decision Lens

Apply in the organisation

- Recalculate base years when policy triggers
- Reconcile initiatives to inventory sources
- Track data-quality improvement
- Investigate unexpected favourable variance

Avoid the trap

- Intensity can improve while absolute emissions rise
- Grid decarbonisation is not an internal project
- Method changes can create artificial trends
- Targets do not prove achievement

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Audit findings, closing meeting and follow-up

1

Conformity

Evidence demonstrates that the stated criterion is met.

2

Nonconformity

Objective evidence shows a requirement is not fulfilled.

3

Observation or OFI

A risk or improvement insight that is not mislabelled as a requirement breach.

4

Closure

Correction addresses the immediate issue; corrective action addresses cause and effectiveness.

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Decision Lens

Apply in the organisation

- Write criterion-evidence-evaluation findings
- Calibrate severity before closing
- Invite factual clarification without negotiating facts
- Verify implementation and effectiveness separately

Avoid the trap

- Do not hide requirements inside recommendations
- A missing record is not automatically a wrong result
- Root cause is not repeat the finding
- Closing evidence can be superficial

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Activities · Topic 4

Activity 9

- Evaluate Uncertainty and Product Benchmark Performance
- K6 · K7 · A5 · A6 · LO4
- 45 minutes

Activity 10

- Close the Audit and Issue a Verification-Ready Report
- K1 · K8 · A1 · A5 · A6 · A8 · LO1 · LO4
- 40 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

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K6 · K7 · A5 · A6 · LO4 · 45 minutes

Evaluate Uncertainty and Product Benchmark Performance

ACTIVITY 9

Learners test whether uncertainty or weak comparability could reverse HarbourLight's proposed lower-carbon product claim.

You'll build

An uncertainty register, sensitivity analysis, benchmark comparability screen, normalised comparison and claim recommendation.

Tools: Product model · uncertainty ranges · fictional EPD extracts · benchmarking workbook

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K6 · K7 · A5 · A6 · LO4 · CONCEPT VIEW

Activity 9 · Evidence Model

1. **Uncertainty**
   Data, factor and model view

2. **Sensitivity**
   Decision-changing inputs

3. **Benchmarks**
   Comparable performance basis

4. **Claim**
   Evidence-bounded conclusion

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ACTIVITY 9 · ACCEPTANCE CHECK

Evaluate Uncertainty and Product Benchmark Performance

✅ Test it

All material uncertainty classes are addressed; sensitivity links to decisions; only genuinely comparable benchmarks influence the conclusion; claim wording matches evidence and limitations.

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K1 · K8 · A1 · A5 · A6 · A8 · LO1 · LO4 · 40 minutes

Close the Audit and Issue a Verification-Ready Report

ACTIVITY 10

Learners complete the HarbourLight capstone with a closing meeting, audit report and follow-up plan.

You'll build

A calibrated finding register, closing presentation, internal-audit report, corrective-action plan and follow-up test schedule.

Tools: Activities 1-9 outputs · finding rubric · report template · management role cards

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K1 · K8 · A1 · A5 · A6 · A8 · LO1 · LO4 · CONCEPT VIEW

Activity 10 · Evidence Model

1 **Findings**
Criterion and consequence

2 **Closing**
Balanced communication

3 **Report**
Scope-accurate conclusion

4 **Follow-up**
Corrective-action effectiveness

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ACTIVITY 10 · ACCEPTANCE CHECK

Close the Audit and Issue a Verification-Ready Report

✅ Test it

Every conclusion is evidence-based; the report is balanced and scope-accurate; disagreement is transparent; corrective action addresses cause; follow-up tests effectiveness rather than document completion alone.

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K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4 · TOPIC CHECK

Recap · Calculations, Uncertainty and Performance

1

Explain

Use compatible units, appropriate factors, declared GWP basis and transparent allocation.

2

Apply

Determine material uncertainty factors and compare the Lumina 400 with credible, method-aligned benchmarks.

3

Demonstrate

Consolidate evidence, communicate balanced conclusions and verify corrective action for the ISO 14064 internal audit.

4

Evidence

Retain the completed activity artefacts and review feedback.

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WRAP-UP

From GHG Evidence to Credible Audit Conclusions

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COURSE OUTCOMES

What You Achieved

1

LO1

Develop process mappings and document GHG accounting reports aligned to ISO 14064 and international verification guidelines.

2

LO2

Define lifecycle boundaries and determine emission sources to identify carbon reduction opportunities in line with assessment principles.

3

LO3

Construct lifecycle inventory data for carbon footprint assessments using product configurations and scoped emission categories.

4

LO4

Determine uncertainty factors and evaluate product carbon performance against industry benchmarks.

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END-TO-END CONCEPT MODEL

The Complete ISO 14064 Internal Audit Cycle

1

Criteria + scope

Declared standards, boundaries, users and assurance intent.

2

Risk plan

Material claims, evidence gaps, sampling and independence.

3

Evidence + findings

Corroboration, recalculation and criterion-based evaluation.

4

Improvement loop

Correction, corrective action and effectiveness follow-up.

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https://lms-tms.tertiaryinfotech.com/

Courseware and Assessment on the LMS

1

Download

Get the learner slides PDF and Learner Guide.

2

Activities

Open the complete Activities folder and working files.

3

Assessment

Complete the WA and Case Study during the scheduled window.

4

Submit

Upload completed answers and confirm successful submission.

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FINAL REMINDER

Assessment

WA

- 8 questions
- K1-K8
- 1 hour · open book

Case Study

- 4 questions
- A1-A8
- 1 hour · open book

Complete

- Assessment attendance
- Submit on LMS
- Sign summary record

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ON ASSESSMENT DAY · CONTROL POINTS

Assessment Readiness

1

Attendance

TRAQOM and assessment attendance are complete.

2

Instruments

Written Assessment and Case Study are completed individually.

3

Submission

Answers are uploaded to the LMS within the assessment window.

4

Record

The Assessment Summary Record is reviewed and signed.

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Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

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THANK YOU

Follow the evidence. Challenge the claim. Improve the system.

Use ISO 14064 principles and disciplined audit judgement to make GHG decisions credible.

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